Supplemental Material: Entangled Matter-waves for Quantum Enhanced Sensing

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CONTENTS

I. NUMERICAL SIMULATION PARAMETERS

For the two relevant atomic levels we consider the D_2 cycling transition, with a transition wavelength of 780 nm and a decay rate of $\gamma = 2\pi \times 6.066$ MHz [1]. The single relevant mode of the cavity couples to the atoms with a single-atom coupling strength of $g = 2\pi \times 0.5$ MHz and decays at a rate $\kappa = 2\pi \times 0.055$ MHz. The atom-pump detuning is $\Delta_a = 2\pi \times 150$ MHz with a pump coupling rate of $\eta = 2\pi \times 10$ MHz, corresponding to the rate at which pump photons make it through the mirror. The atoms are accelerated to have a mean momentum of $p_0 = 15\hbar k$, which provides an energy splitting of $\omega_z = -2\pi \times 0.44$ MHz, which matches Ref. [2] to an order of magnitude.

II. THE DIPOLE RADIATION PATTERN

The dipole radiation pattern is calculated by considering a specific dipole radiation pattern. As an example, in simulations we consider the rubidium dipole allowed transition from $F = 2, m_F = 2$ to $F = 3, m_f = 3$ and tracing out the x and y degrees of freedom [3, 4]. This yields

$$\hat{d}_\mu(\theta) = \sqrt{\mathcal{N}(\theta)} \exp[-ik\hat{z}_\mu \cos(\theta)], \quad \mathcal{N}(\theta) = \frac{3}{8}(1 + \cos(\theta)^2), \tag{1}$$

where θ is the angle between the emitted photon and the positive z-axis [3, 4]. If we had considered states with another angular momentum difference, this would lead to a different dipole radiation pattern. The full term in the Lindblad equation is then integrated over θ:

$$\frac{\hat{\mathcal{D}}[\hat{L}_{\mathrm{sp},\mu}]\hat{\rho}}{\gamma} = \int_{-1}^{+1} \Big[\mathcal{N}(u) \exp(-ik\hat{z}_\mu u)\hat{\sigma}_\mu^- \hat{\rho}\hat{\sigma}_\mu^+ \exp(+ik\hat{z}_\mu u) - \{\hat{\sigma}_\mu^+ \hat{\sigma}_\mu^-, \hat{\rho}\}/2 \Big] du. \tag{2}$$

III. ADIABATIC ELIMINATION OF THE EXCITED STATE

From Eq. (1) in the main text, we eliminate the excited state based on the assumption $|\Delta_a| \gg \sqrt{N}g$, with $\Delta_a \equiv \omega_a - \omega_p$. We assume that all atoms start in the ground state, i.e. $\hat{\rho}_{\mathrm{org}} = \hat{\rho}_{k,c} \otimes \hat{\rho}_a$ where

$$\hat{\rho}_a = \bigotimes_{\mu=1}^{N} |g\rangle_\mu \langle g|_\mu . \tag{3}$$

Now, we because the excited state is far detuned, any frequencies (timescales) are much bigger (faster) for excited state dynamics compared to the relevant atomic motion we wish to study. Subject to this assumptions, we take make the approximation that $(\partial/\partial t) \langle e|_\mu \hat{\rho}_a |g\rangle_\mu \approx 0$ for all relevant timescales. Now, using the Heisenberg picture of evolution:

$$\frac{\partial \hat{A}}{\partial t} = \frac{i}{\hbar}\left[\hat{H}, \hat{A}\right] + \sum_j \hat{\mathcal{D}}^\dagger[\hat{L}_j]\hat{A}. \tag{4}$$

this approximation is equivalent to the condition that

$$(\partial/\partial t)\hat{\sigma}_\mu^- \approx 0. \tag{5}$$

Therefore, we can directly solve for this condition in order to find an effective replacement for the atom internal state observables:

$$(\partial/\partial t)\hat{\sigma}_\mu^- = -i\Delta_a \hat{\sigma}_\mu^- + ig\cos(k\hat{z}_\mu)\hat{a}\hat{\sigma}_\mu^z - \frac{\gamma}{2}\hat{\sigma}_\mu^- \approx 0, \tag{6}$$

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and therefore

$$\hat{\sigma}^-_\mu \approx \frac{g}{2\Delta_a - i\gamma}\cos(k\hat{z}_\mu)\hat{a}\hat{\sigma}^z_\mu. \tag{7}$$

Now, because we have assumed Eq. 3, we may replace $\hat{\sigma}^z_\mu \to -\left|g\right\rangle_\mu \left\langle g\right|_\mu$, where the atoms are always in the ground state so $\left|g\right\rangle_\mu \left\langle g\right|_\mu = 1$, yielding

$$\hat{\sigma}^-_\mu \approx \frac{-g}{2\Delta_a - i\gamma}\cos(k\hat{z}_\mu)\hat{a}. \tag{8}$$

To reconstruct the correct Hamiltonian and jump operators, we consider the time evolution of the cavity operator $\hat{a}$, and of the momentum operator $\hat{p}_\mu$:

$$\begin{aligned}\frac{\partial}{\partial t}\hat{a} &= \frac{i}{\hbar}[\hat{H}_{\text{rot}}, \hat{a}] + \hat{\mathcal{D}}^\dagger[\hat{L}_{\text{cav}}]\hat{a} \\ \frac{\partial}{\partial t}\hat{p}_\mu &= \frac{i}{\hbar}[\hat{H}_{\text{rot}}, \hat{p}_\mu] + \hat{\mathcal{D}}^\dagger[\hat{L}_{\text{sp},\mu}]\hat{p}_\mu\end{aligned} \tag{9}$$

where $\hat{\mathcal{D}}^\dagger[\hat{L}_{\text{sp},\mu}]\hat{p}_\mu$ includes integration about θ as in Eq. (2). We replace the action of $\hat{\sigma}^-_\mu$ with the right hand side of Eq. 8 to find the atom-field Hamiltonian and jump operators

$$\begin{aligned}\hat{H}_{\text{af}} &= \hbar\Delta_c'\hat{a}^\dagger\hat{a} + \hbar\eta\left(\hat{a} + \hat{a}^\dagger\right) \\ &\quad + \sum_{\mu=1}^{N}\left(\frac{\left(\hat{p}_\mu - p_0\right)^2}{2m} - \hbar U_0\ \hat{a}^\dagger\hat{a}\cos(2k\hat{z}_\mu)\right), \\ \hat{L}_{\text{cav}} &= \sqrt{\kappa}\hat{a}, \quad \hat{L}_{\text{rec},\mu}(\theta) = \sqrt{2\gamma U_0/\Delta_a}\ \hat{d}_\mu(\theta)\hat{a}\cos(k\hat{z}_\mu),\end{aligned} \tag{10}$$

with $\Delta_c' = \Delta_c - U_0 N$ being the dressed cavity detuning. The state may now be found by

$$\text{Tr}_a(\hat{\rho}_{\text{org}}) = \hat{\rho}_{k,c}\,\text{Tr}_a(\hat{\rho}_a) = \hat{\rho}_{k,c}, \tag{11}$$

where we have again used the assumption that Eq. 3 is true for all relevant timescales.

IV. DISCRETIZED MOMENTUM SPACE

We note that the single particle jump operator does not fit conveniently in this formalism, but later when considering squeezing we treat single particle decoherence explicitly. The entire Hamiltonian now contains only collective momentum physics, which we second quantize via

$$\begin{aligned}&\int \hat{\psi}^\dagger(p)\hat{\psi}(p)dp = \\ &\sum_n \int_{(2n-1)\hbar k}^{(2n+1)\hbar k} \hat{\psi}^\dagger(p)\hat{\psi}(p)dp = \\ &\sum_n \hat{b}_n^\dagger\hat{b}_n,\end{aligned} \tag{12}$$

so that each sum over all particles is replaced by an integral over an operator valued measure. We also impose that $\sum_n \hat{b}_n^\dagger\hat{b}_n = N$, ie there are N total atoms. We additionally assume the particles have near zero-momentum width, so that

$$\begin{aligned}\hat{b}_n^\dagger\hat{b}_n &= \int_{(2n-1)\hbar k}^{(2n+1)\hbar k} \hat{\psi}^\dagger(p)\hat{\psi}(p)dp \\ &\approx \int_{(2n-\epsilon)\hbar k}^{(2n+\epsilon)\hbar k} \hat{\psi}^\dagger(p)\hat{\psi}(p)dp,\end{aligned} \tag{13}$$

for $\epsilon \ll 1$. This means that $2n\hbar k + p \approx 2n\hbar k$ in terms like the kinetic energy. Going term by term, this yields;

$$\begin{aligned}\sum_j \frac{\left(\hat{p}_j - p_0\right)^2}{2\hbar m} &= \int_{-\infty}^{+\infty} \hat{\psi}^\dagger(p)\frac{\left(p - p_0\right)^2}{2\hbar m}\hat{\psi}(p)dp \\ &\approx \sum_n \int_{(2n-\epsilon)\hbar k}^{(2n+\epsilon)\hbar k} \hat{\psi}^\dagger(p)\frac{\left(2n\hbar k - p_0\right)^2}{2\hbar m}\hat{\psi}(p)dp \\ &= \sum_n \hat{b}_n^\dagger\hat{b}_n K_n,\end{aligned} \tag{14}$$

where $K_n = \left[(n - n_0)2\hbar k\right]^2/2\hbar m$ is the kinetic energy for the n^{th} momentum state and n_0 is the mean momentum state such that $p_0 = n_0 2\hbar k$. Each of the trigonometric operators is simply expressed in this formalism as well;

$$\begin{aligned}\sum_j \mathrm{e}^{\pm i2k\hat{z}_j} &= \sum_n \int_{(2n-1)\hbar k}^{(2n+1)\hbar k} \hat{\psi}^\dagger(p)\mathrm{e}^{\pm i2k\hat{z}}\hat{\psi}(p)dp \\ &= \sum_n \int_{(2n-1)\hbar k}^{(2n+1)\hbar k} \hat{\psi}^\dagger(p \pm 2\hbar k)\hat{\psi}(p)dp \\ &= \sum_n \hat{b}_{n\pm1}^\dagger\hat{b}_n \equiv \sum_n \hat{J}_n^\pm,\end{aligned} \tag{15}$$

which we use to write

$$\hat{C}_n \equiv \frac{\hat{J}_n^+ + \hat{J}_{n+1}^-}{2}, \quad \sum_j \cos(2k\hat{z}_j) = \sum_n \hat{C}_n, \tag{16}$$

and

$$\hat{S}_n \equiv \frac{\hat{J}_n^+ - \hat{J}_{n+1}^-}{2i}, \quad \sum_j \sin(2k\hat{z}_j) = \sum_n \hat{S}_n. \tag{17}$$

We note that $[\hat{C}_n, \hat{S}_n] = \frac{i}{2}(\hat{b}_{n+1}^\dagger\hat{b}_{n+1} - \hat{b}_n^\dagger\hat{b}_n)$, following from an $\mathfrak{su}(2)$ commutation relation. This means the Hamiltonian after displacement is

$$\begin{aligned}\frac{\tilde{\hat{H}}_{\text{af}}}{\hbar} &= \Delta_c'\tilde{a}^\dagger\tilde{a} + \sum_n\left(K_n\hat{b}_n^\dagger\hat{b}_n - U_0|\beta|^2\hat{C}_n\right) \\ &\quad - U_0\left(\tilde{a}^\dagger\tilde{a} + \tilde{a}^\dagger\beta + \beta^*\tilde{a}\right)\sum_n \hat{C}_n.\end{aligned} \tag{18}$$

V. SCATTERED PHOTONS IN THE CAVITY

Now, we eliminate the cavity degrees of freedom from Eq. (3). As noted in the main text, the cavity response to the pump and atomic motion can be separated into two components. The pump leads to a displacement of the cavity mode by amount $\beta \equiv \eta/(-\Delta_c' + i\kappa/2)$, yielding

$$\hat{D}_1(\beta)^\dagger \hat{a} \hat{D}_1(\beta) = \beta + \hat{a}, \tag{19}$$

where $\hat{D}_1(\beta) \equiv \exp\left(\hat{a}^\dagger\beta - \beta^*\hat{a}\right)$ is the displacement operator. Now β is the coherent injected field amplitude. The Hamiltonian in this displaced picture is correspondingly

$$\begin{aligned}\frac{\tilde{H}_{\text{af}}}{\hbar} \equiv& \Delta_c'\hat{a}^\dagger\hat{a} + \Delta_c'|\beta|^2 + \sum_n K_n \hat{b}_n^\dagger \hat{b}_n - U_0\hat{C}_n(\beta^* + \hat{a}^\dagger)(\beta + \hat{a}) \\ =&(\Delta_c' - U_0\sum_n \hat{C}_n)\hat{a}^\dagger\hat{a} + \Delta_c'|\beta|^2 + \sum_n K_n \hat{b}_n^\dagger \hat{b}_n - U_0|\beta|^2\hat{C}_n - U_0\beta^*\hat{C}_n\hat{a} - U_0\beta\hat{C}_n\hat{a}^\dagger.\end{aligned} \tag{20}$$

The energy due to the classical field, $\Delta_c'|\beta|^2$ is just a constant real number so it will only add a global phase and we can drop it. Now, the Hamiltonian has three components. First, $\tilde{H}_k \equiv \hbar\sum_n \left(K_n\hat{b}_n^\dagger\hat{b}_n - U_0|\beta|^2\hat{C}_n\right)$ is the momentum only Hamiltonian, where now the classical injected field drives a cosine interaction with the atoms. Second, $\tilde{H}_c = \hbar\Delta_c'\tilde{a}^\dagger\tilde{a}$ is the photon only Hamiltonian. Lastly, the interaction Hamiltonian is $\tilde{H}_{\text{int}} = -\hbar U_0\sum_n \hat{C}_n(\hat{a}^\dagger\hat{a} - \beta^*\hat{a} - \beta\hat{a}^\dagger)$, where the atomic motion modifies the frequency of the photons through the first term, and scatters photons in to and out of the coherent field through the second and third terms. Now, these photons start in a displaced vacuum about β and, as time goes on, only acquire non-trivial dynamics through the coupling to atomic motion, coupled by a small frequency U_0. Therefore, we can adiabatically elminate the photons. To do this, we use one more displacement

$$\hat{D}_2(\hat{\alpha}(t)) \equiv \exp\left(\tilde{a}^\dagger\hat{\alpha}(t) - \hat{\alpha}(t)^\dagger\tilde{a}\right) \tag{21}$$

where $\hat{\alpha}(t)$ is a position/momentum operator, and we solve for the new master equation. This displacement replaces the photonic equations of motion with the equations of motion for the operators that drive them, in this case $\hat{J}_n^\pm$ representing the two photon recoil. We solve for the equations of motion of $\hat{\alpha}(t)$ following Ref. [5];

$$\frac{\partial}{\partial t}\hat{\alpha}(t) = -\frac{i}{\hbar}\left[\hat{H}_k, \hat{\alpha}(t)\right] - i\left(\Delta_c' - U_0\sum_n \hat{C}_n\right)\hat{\alpha}(t) - \frac{\kappa}{2}\hat{\alpha}(t) + iU_0\beta\sum_n \hat{C}_n. \tag{22}$$

Here, we observe that $\Delta_c' - U_0\sum_n \hat{C}_n = \Delta_c'(1 - \frac{U_0}{\Delta_c'}\sum_n \hat{C}_n) \approx \Delta_c'$. Dropping this term is equivalent to dropping the next order of modulation, i.e. the frequency shift of the quantum degrees of freedom $\hat{\alpha}(t)$ by the atoms. In other words, we only solve for the central cavity peak interacting with the atoms (first order), the frequency modulation of the central cavity peak (second order), and the interaction of the atoms with that peak (third order), and not the fourth order modulation of these scattered photons–which is the same order as the modulation due to spontaneous emission that we already dropped. All together, we must solve

$$\frac{\partial}{\partial t}\hat{\alpha}(t) = -i\left[\sum_n K_n\hat{b}_n^\dagger\hat{b}_n, \hat{\alpha}(t)\right] - i\left(\Delta_c' - i\frac{\kappa}{2}\right)\hat{\alpha}(t) + iU_0\beta\sum_n \hat{C}_n. \tag{23}$$

We can make the ansatz that

$$\hat{\alpha}(t) = \beta\sum_n \left(A_n^+(t)\hat{J}_n^+ + A_n^-(t)\hat{J}_n^-\right)/2, \tag{24}$$

where the amplitude $A_n^\pm(t)$ carries the time and momentum dependence for photon absorption in the $\pm z$ direction, and $\hat{J}_n^\pm$ is the corresponding momentum kick direction in that direction. We note that

$$\left[\sum_n K_n\hat{b}_n^\dagger\hat{b}_n, \sum_m \hat{J}_m^\pm\right] = \sum_n (K_{n\pm1} - K_n)\hat{J}_n^\pm. \tag{25}$$

This ansatz yields two uncoupled differential equations;

$$\dot{A}_n^\pm(t) = -i(K_{n\pm1} - K_n)\ A_n^\pm(t) - i\left(\Delta_c' - i\frac{\kappa}{2}\right)A_n^\pm(t) + iU_0, \tag{26}$$

which have solutions

$$A_n^{\pm} \approx \frac{U_0}{\Delta_c' + (K_{n\pm1} - K_n) - i\kappa/2}, \tag{27}$$

where we've dropped the initial time dependent decay from the pump being turned on. Now, we note that $[\hat{\alpha}, \sum_n \hat{C}_n] = 0$ and substitute $\hat{\alpha}$ this back into the Hamiltonian;

$$\begin{aligned} \frac{\hat{H}_a}{\hbar} &= \sum_n \left(K_n \hat{b}_n^\dagger \hat{b}_n - U_0|\beta|^2 \hat{C}_n - \frac{U_0}{2}\left[\beta\hat{\alpha}^\dagger + \beta^*\hat{\alpha}\right]\hat{C}_n \right) \\ &= \sum_n \left(K_n \hat{b}_n^\dagger \hat{b}_n - U_0|\beta|^2 \hat{C}_n - \frac{U_0|\beta|^2}{2} \sum_m \Re\left[A_m^+ \hat{J}_m^+ + A_m^- \hat{J}_m^-\right]\hat{C}_n \right). \end{aligned} \tag{28}$$

Now, the atoms interacts with the injected field, of intensity $|\beta|^2$, and the atoms feel the effects of their own motion reflected back on them through this second term. In order to make physical sense of this second term, we write $\hat{J}_m^{\pm}$ in the real (hermitian) and imaginary (anti-hermitian) components:

$$\hat{J}_m^+ = \hat{C}_m + i\hat{S}_m, \quad \text{and} \quad \hat{J}_m^- = \hat{C}_{m-1} + i\hat{S}_{m-1}. \tag{29}$$

We also note that $\Re(YZ) = \Re(Y)\Re(Z) - \Im(Y)\Im(Z)$. Therefore

$$\begin{aligned} \sum_m \Re\left[A_m^+ \hat{J}_m^+ + A_m^- \hat{J}_m^-\right] &= \sum_m \left[\Re\left(A_m^+\right)\hat{C}_m + \Re\left(A_m^-\right)\hat{C}_{m-1} - \Im\left(A_m^+\right)\hat{S}_m - \Im\left(A_m^-\right)\hat{S}_{m-1}\right] \\ &= \sum_m \left[\Re\left(A_m^+ + A_{m+1}^-\right)\hat{C}_m - \Im\left(A_m^+ - A_{m+1}^-\right)\hat{S}_m\right]. \end{aligned} \tag{30}$$

This yields

$$\frac{\hat{H}_a}{\hbar} = \sum_n \left(K_n \hat{b}_n^\dagger \hat{b}_n - U_0|\beta|^2 \hat{C}_n - \sum_m \left[\chi_m \hat{C}_m + \zeta_m \hat{S}_m\right]\hat{C}_n \right). \tag{31}$$

where

$$\chi_n \equiv \frac{U_0|\beta|^2}{2}\Re(A_n^+ + A_{n+1}^-), \;\; \zeta_n \equiv \frac{U_0|\beta|^2}{2}\Im(A_{n+1}^- - A_n^+) \tag{32}$$

are the rates for the cosine and sine non-linear interactions. This adiabatic elimination also yields the collective jump operator $\hat{L}_c = \sqrt{\kappa|\beta|^2}\, \sum_n \left(A_n^+ \hat{J}_n^+ + A_n^- \hat{J}_n^-\right)/2$.

VI. TWO LEVEL MAPPING

Now, we restrict ourselves to just $n = 0, +1$ (the $0\hbar k$ and $+2\hbar k$ states). By restricting to these two levels, we are left with only two-level operators;

$$\begin{aligned} \hat{J}_x &\equiv \hat{C}_0 = \frac{1}{2}\left(\hat{J}_0^+ + \hat{J}_1^-\right), \\ \hat{J}_y &\equiv \hat{S}_0 = \frac{1}{2i}\left(\hat{J}_0^+ - \hat{J}_1^-\right) \\ \hat{J}_z &\equiv -i[\hat{J}_x, \hat{J}_y] = \frac{1}{2}(\hat{b}_1^\dagger \hat{b}_1 - \hat{b}_0^\dagger \hat{b}_0) \end{aligned} \tag{33}$$

These two momentum states are separated by a kinetic energy $\hbar\omega_z \equiv \hbar(K_{+1} - K_0)$.

Now, we use the following:

$$\hat{b}_n \left|\psi\right> = 0, \quad \text{and} \quad \hat{J}_n^{\pm}\left|\psi\right> = 0, \quad \text{for} \quad n \neq 0, +1. \tag{34}$$

This means

$$\begin{aligned} \frac{\hat{H}_a}{\hbar} =& K_0 \hat{b}_0^\dagger \hat{b}_0 + K_1 \hat{b}_1^\dagger \hat{b}_1 - U_0|\beta|^2 \hat{C}_0 \\ & - \chi_0 \hat{C}_0^2 + \zeta_0(\hat{S}_0\hat{C}_0 + \hat{C}_0\hat{S}_0) \\ & - \hat{B}_1 - \hat{B}_2 - \hat{B}_3 - \hat{B}_4 \end{aligned} \tag{35}$$

where the terms labeled $\hat{B}_j$ for $j = 1, 2, 3, 4$ break the

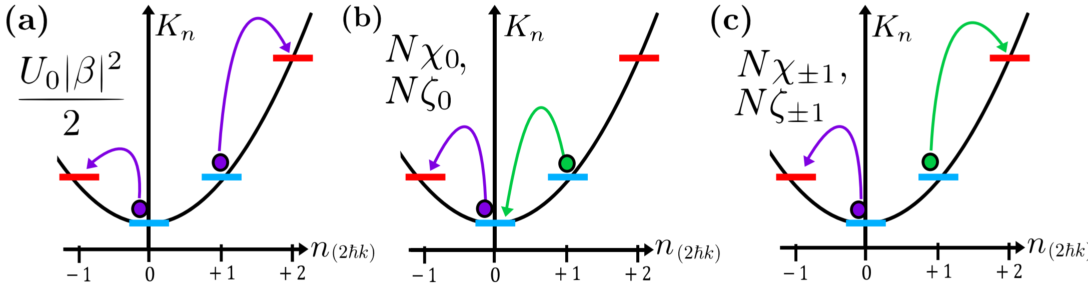

FIG. 1. (a) Single atom processes which break the two-level assumption, due to Bragg scattering off of the optical lattice. (b) and (c) Multi-atom process which break the two-level approximation. (b) An example transition which occurs with rate $N\chi_0$ or $N\zeta_0$, where one atom (purple) leaves the two-level manifold but the other atom (green) does not. This is the multi-atom process which pre-dominantly breaks the two-level approximation. (c) An example transition which occurs with rate $N\chi_1$ or $N\zeta_1$, where both atom (purple and green) leave the two-level manifold. This the multi-atom process has a higher energy barrier than if one atom stayed in the manifold.

two-level approximation;

$$\begin{aligned}
\hat{B}_1 \equiv& U_0|\beta|^2\left(\hat{J}_0^- + \hat{J}_1^+\right)/2\\
\hat{B}_2 \equiv& \left[\chi_0\hat{C}_0 + \zeta_0\hat{S}_0\right]\left(\hat{J}_0^- + \hat{J}_1^+\right)/2\\
\hat{B}_3 \equiv& \left[\chi_{-1}\hat{J}_0^- + \chi_1\hat{J}_1^+ + \zeta_{-1}\hat{J}_0^- + \zeta_1\hat{J}_1^+\right]\hat{C}_0\\
\hat{B}_4 \equiv& \left[\chi_{-1}\hat{J}_0^- + \chi_1\hat{J}_1^+ + \zeta_{-1}\hat{J}_0^- + \zeta_1\hat{J}_1^+\right](\hat{J}_0^- + \hat{J}_1^+)/2
\end{aligned} \tag{36}$$

if we only keep the terms that keep us in the two-level manifold we get, and gauge transform away terms proportional to $(\hat{b}_0^\dagger\hat{b}_0 + \hat{b}_1^\dagger\hat{b}_1)$, then we find

$$\begin{aligned}
\frac{\hat{H}_{\text{eff}}}{\hbar} =& \omega_z\hat{J}_z - U_0|\beta|^2\hat{J}_x - \chi_0\hat{J}_x^2 - \frac{\zeta_0}{2}\left(\hat{J}_x\hat{J}_y + \hat{J}_y\hat{J}_x\right),\\
\hat{L}_c =& \sqrt{\kappa|\beta|^2/4}\,(A_0^+\hat{J}^+ + A_{+1}^-\hat{J}^-),
\end{aligned} \tag{37}$$

where we drop the subscript in the main text: $\chi_0 \to \chi$ and $\zeta_0 \to \zeta$.

A. Restrictions on the Two-Level Mapping

Separating these two momentum states requires that none of the terms $\hat{B}_j$ drive atoms out of this two level manifold. These terms amount to Bragg scattering, $\hat{B}_1$, and the nonlinear cosine-cosine and cosine-sine interactions, $\hat{B}_2$, $\hat{B}_3$, and $\hat{B}_4$.

The first term, Bragg scattering, could break the two level approximation by allowing the following single particle transition:

$$|0\hbar k\rangle \to |-2\hbar k\rangle \quad \text{or} \quad |+2\hbar k\rangle \to |+4\hbar k\rangle \tag{38}$$

through the transition elements;

$$\begin{aligned}
\langle n=-1|\,\hat{B}_1\,|n=0\rangle =& \frac{U_0|\beta|^2}{2}\\
\langle n=+2|\,\hat{B}_1\,|n=+1\rangle =& \frac{U_0|\beta|^2}{2},
\end{aligned} \tag{39}$$

shown schematically in Fig. 1 (a). These single particle terms come with the energy condition:

$$|K_{-2} - K_0|, \quad \text{and} \quad |K_{+2} - K_{+1}|. \tag{40}$$

Using this, we can consider the effective Rabi-oscillation between $0\hbar k$ and $-2\hbar k$ or $+2\hbar k$ and $+4\hbar k$, which yields the energy condition the maximum probability of excitation [6]:

$$\frac{\left(\frac{U_0|\beta|^2}{2}\right)^2}{\left(\frac{U_0|\beta|^2}{2}\right)^2 + |\omega_k|^2} \ll 1 \tag{41}$$

for $\omega_k = \min\left(|K_{-1} - K_0|,\ |K_{+2} - K_{+1}|\right)$ being the smaller of the two energy gaps. This yields the condition that

$$U_0|\beta|^2 \ll 2\min\left(\ |K_{-1} - K_0|,\ |K_{+2} - K_{+1}|\ \right). \tag{42}$$

In the limit that that atoms are fast moving, $p_0 \gg 2\hbar k$, this yields

$$U_0|\beta|^2 \ll \frac{4kp_0}{m}. \tag{43}$$

which means that the standing field will also have low probability of driving $|0\hbar k\rangle \leftrightarrow |+2\hbar k\rangle$ as well.

The second set of terms, $\hat{B}_2$, $\hat{B}_3$, and $\hat{B}_4$ correspond to non-linear interactions. These drive two atom transitions, such as the two examples in Fig. 1 (b) and (c) and therefore have many more ways to break the two-level approximation. For two atomic states labeled by the pair $(n,m) = \mathcal{S}(|n\rangle \otimes |m\rangle)$ with the symmetrizer $\mathcal{S}$ and momentum states $|n\rangle = |n2\hbar k\rangle$, the transitions which break the two level approximation are

$$\begin{array}{ll}
(0,0) \to (-1,-1) & 2(K_{-1} - K_0)\\
(0,0) \to (-1,+1) & K_{+1} + K_{-1} - 2K_0\\
(0,+1) \to (-1,0) & K_{-1} - K_{+1}\\
(0,+1) \to (-1,+2) & K_{-1} + K_{+2} - K_0 - K_{+1}\\
(0,+1) \to (+1,+2) & K_{+2} - K_0\\
(+1,+1) \to (0,+2) & K_0 + K_{+2} - 2K_{+1}\\
(+1,+1) \to (+2,+2) & 2(K_{+2} - K_{+1})
\end{array} \tag{44}$$

Each of these have a corresponding matrix element, such as in the $\hat{B}_1$ case. However, the transitions $(0,0) \to (-2,+2)$ and $(2,2) \to (0,+4)$ from will always have the lowest energy cost with $K_{+1}+K_{-1}-2K_0 = K_0+K_{+2}-2K_{+1} = 8\hbar^2k^2/2\hbar m$. Furthermore, we care about maximizing χ_0 and ζ_0 to optimize squeezing on this two level manifold, so we only bound these two rates. In principle, this process provides bounds for every rate appearing in $\hat{B}_j$. This gives:

$$N|\chi_0|,\ N|\zeta_0| \ll \frac{8\hbar^2k^2}{2\hbar m} \tag{45}$$

So, these two restrictions together are

$$|U_0||\beta|^2 \ll \frac{4kp_0}{m}, \quad \text{and} \quad N|\chi_0|,\ N|\zeta_0| \ll \frac{8\hbar^2k^2}{2\hbar m}. \tag{46}$$

B. Single Particle Decay

In the two-state approximation for momentum, the spontaneous emission breaks permutational symmetry. To calculate the final jump operators in the two mode picture, we preemptively define single particle pseudo-spin operators;

$$\exp(\pm i2k\hat{z}_\mu) \approx \hat{s}^{\pm}_\mu \equiv |{+2\hbar k}\rangle_\mu \langle 0\hbar k|_\mu\,, \tag{47}$$

Furthermore, single particle spontaneous emission doesn't stay on the momentum grid like coherent dynamics or collective decay. To treat this, we assume that

$$\mathcal{N}(u) = A(\delta(u+1)+\delta(u-1)) + B\delta(u), \tag{48}$$

for $u = \cos(\theta)$ and where A and B are chosen such that the first and second moments of the momentum-recoil matches the continuous case [3, 4]. With two free variables, A and B, this is the best we can do. This yields $A = 1/5$ and $B = 3/5$, which means the original spontaneous emission operator in Eq. (2) becomes

$$\begin{aligned}\hat{\mathcal{D}}[\hat{L}_\mu]\hat{\rho} =& \frac{\gamma}{5}\left(\exp(-ik\hat{z}_\mu)\hat{\sigma}^-_\mu\hat{\rho}\hat{\sigma}^+_\mu\exp(+ik\hat{z}_\mu)\right) + ...\\ &+\frac{\gamma}{5}\left(\exp(+ik\hat{z}_\mu)\hat{\sigma}^-_\mu\hat{\rho}\hat{\sigma}^+_\mu\exp(-ik\hat{z}_\mu)\right) + ...\\ &+\frac{3\gamma}{5}\left(\hat{\sigma}^-_\mu\hat{\rho}\hat{\sigma}^+_\mu\right) - \frac{\gamma}{2}\{\hat{\sigma}^+_\mu\hat{\sigma}^-_\mu,\hat{\rho}\}\\ =&\hat{\mathcal{D}}[\hat{L}^{+1}_\mu]\hat{\rho} + \hat{\mathcal{D}}[\hat{L}^{-1}_\mu]\hat{\rho} + \hat{\mathcal{D}}[\hat{L}^0_\mu]\hat{\rho}.\end{aligned} \tag{49}$$

where

$$\hat{L}^{\pm1}_\mu \equiv \sqrt{\frac{\gamma}{5}}\exp(\pm ik\hat{z}_\mu)\hat{\sigma}^-_\mu\,, \quad \text{and} \quad \hat{L}^0_\mu \equiv \sqrt{\frac{3\gamma}{5}}\hat{\sigma}^-_\mu. \tag{50}$$

After eliminating the excited state we find

$$\tilde{\tilde{L}}_{\mathrm{rec},\mu}(\theta) = \sqrt{2\gamma U_0/\varDelta_a}\ \hat{d}_\mu(\theta)(\beta+\tilde{a})\cos(k\hat{z}_\mu), \tag{51}$$

but we assume that spontaneous emission is only due to absorption from the coherent field, so that $(\beta + \tilde{a})\cos(k\hat{z}_\mu) \approx \beta\cos(k\hat{z}_\mu)$. This yields

$$\hat{L}_\mu \equiv \sqrt{\frac{\gamma U_0|\beta|^2\mathcal{N}(u)}{(\varDelta_a/2)}}\ \cos(k\hat{z}_\mu)\exp(-ik\hat{z}_\mu u) \tag{52}$$

which, in the discrete momentum family, becomes

$$\begin{aligned}\hat{L}^{\pm1}_\mu &\equiv \sqrt{\frac{2\gamma U_0|\beta|^2}{5\varDelta_a}}\ \cos(k\hat{z}_\mu)\exp(\pm ik\hat{z}_\mu),\\ \hat{L}^0_\mu &\equiv \sqrt{\frac{6|\beta|^2\gamma U_0}{5\varDelta_a}}\ \cos(k\hat{z}_\mu).\end{aligned} \tag{53}$$

Here, we note that $\hat{L}^0_\mu$ acting on the state $0\hbar k$ or $+2\hbar k$ always leads to atom loss from the 2-level manifold. $\hat{L}^{\pm1}$, however, does not:

$$\begin{aligned}\cos(k\hat{z}_\mu)\exp(\pm ik\hat{z}_\mu) =& \frac{1}{2}\left(\exp(\pm i2k\hat{z}_\mu)+1\right)\\ =&\frac{1}{2}\left(\hat{s}^{\pm}_\mu+1\right)\end{aligned} \tag{54}$$

and, therefore,

$$\hat{L}^{\pm1}_\mu \equiv \sqrt{\frac{\gamma U_0|\beta|^2}{(10\varDelta_a)}}\ (\hat{s}^{\pm}_\mu+1), \quad \hat{L}^0_\mu \equiv \sqrt{\frac{3\gamma U_0|\beta|^2}{(10\varDelta_a)}}\ \hat{\ell}_\mu. \tag{55}$$

where $\hat{\ell}_\mu \approx 2\cos(k\hat{z}_\mu)$ represents atom loss into states like $\left|\pm\frac{1}{2}2\hbar k\right\rangle$. The approximation that an atom in these states is lost is valid for short times because while these states remain relatively un-populated.

VII. OPTIMAL SPIN SQUEEZING PARAMETER

We take the one axis twisting Hamiltonian from the main text, $\hat{H}_{\mathrm{OAT}}$ and ignore the many-body energy gap, $\hat{J}^2$;

$$\hat{H}_{\mathrm{eff}} = \hbar\chi_0\hat{J}_z^2, \quad \hat{L}_\pm \equiv \sqrt{\varGamma_\pm}\ \hat{J}^\pm, \quad \hat{L}_\mu \equiv \sqrt{\gamma_l}\hat{\ell}_j, \tag{56}$$

where

$$\begin{aligned}\chi_0 \equiv& \frac{-\chi}{2}, \quad \varGamma_\pm \equiv \frac{\kappa|\beta|^2|A^\pm|^2}{4},\\ \text{and} \quad &\gamma_l \equiv \frac{\gamma U_0|\beta|^2}{2\varDelta_a},\end{aligned} \tag{57}$$

for particle loss rate γ_l found in Eq. (55). As an aside, we have assumed that all three decay channels lead to particle loss–which strictly over-estimates the total decay.

The squeezing parameter is given by:

$$\xi^2 = \min_\theta \frac{N(\varDelta J^\perp_\theta)^2}{|\langle\mathbf{J}\rangle|^2}. \tag{58}$$

For short squeezing times the average pseudo-spin is in the x-direction, and the length goes as, $\langle \mathbf{J} \rangle = S\mathrm{e}^{-\gamma_l t}$, for $S \equiv N/2$ being the total initial pseuo-spin length. We note that $|\langle \mathbf{J} \rangle|^2/N = S\mathrm{e}^{-2\gamma_l t}/2$.

The minimum over θ is the smallest eigenvalue of the covariance matrix;

$$\mathbf{C} = \begin{pmatrix} \mathrm{var}(\hat{J}_y) & \mathrm{cov}(\hat{J}_y, \hat{J}_z) \\ \mathrm{cov}(\hat{J}_z, \hat{J}_y) & \mathrm{var}(\hat{J}_z) \end{pmatrix} \tag{59}$$

for $\mathrm{cov}(\hat{A}, \hat{B}) \equiv \langle \{\hat{A}, \hat{B}\} \rangle_\psi /2 - \langle \hat{A} \rangle_\psi \langle \hat{B} \rangle_\psi$ and $\mathrm{var}(\hat{A}) = \mathrm{cov}(\hat{A}, \hat{A})$. The state is initially polarized in x and therefore $\langle \hat{J}_y \rangle = \langle \hat{J}_z \rangle = 0$ over short times. The eigenvalues of $\mathbf{C}$ are:

$$V_\pm = \frac{1}{2}\left(\mathcal{A}(t) + \mathcal{B}(t) \pm \sqrt{\left(\mathcal{B}(t) - \mathcal{A}(t)\right)^2 + \mathcal{C}(t)^2} \right), \tag{60}$$

where

$$\begin{aligned} \mathcal{A}(t) = \Delta J_y^2, \quad \mathcal{B}(t) = \Delta J_z^2, \\ \mathcal{C}(t) = \langle \{\hat{J}_y, \hat{J}_z\} \rangle, \end{aligned} \tag{61}$$

For short times, the OAT dynamics can be treated independently of collective decay, and fluctuations from collective emission are added in quadrature to the fluctuations due to squeezing. We first solve for the spin squeezing parameter in the collcetive case, $(\xi_0)^2$, and then add the effects of collective decay back in by modifying ΔJ_z^2, with an additional term for each collective decay channel:

$$\begin{aligned} T(\Gamma_\pm) &= S\tanh(2S\Gamma_\pm t)(1 - \tanh(2S\Gamma_\pm t)) \\ \Delta J_z^2 &\to \Delta J_z^2 + T(\Gamma_+) + T(\Gamma_-), \end{aligned} \tag{62}$$

which requires that $N\Gamma_+ t$, $N\Gamma_- t \ll 1$.

Without collective decay, we find [7]:

$$\begin{aligned} \Delta J_y^2 =& \frac{S\mathrm{e}^{-\gamma_l t}}{2} \\ &+ \frac{S}{2}\left(S - \frac{1}{2}\right)\left(1 - \cos^{(2S-2)}(2\chi_0 t)\right)\mathrm{e}^{-2\gamma_l t} \\ \Delta J_z^2 =& \frac{S\mathrm{e}^{-\gamma_l t}}{2} \\ \langle \{\hat{J}_y, \hat{J}_z\} \rangle =& 2S\left(S - \frac{1}{2}\right)\sin(\chi_0 t)\cos^{(2S-1)}(\chi_0 t)\mathrm{e}^{-2\gamma_l t}. \end{aligned} \tag{63}$$

For intermediate times, we have that $\mathcal{A} \gg \mathcal{B}, \mathcal{C}$, and therefore

$$\begin{aligned} V_- =& \frac{1}{2}\left(\mathcal{A}(t) + \mathcal{B}(t) - \sqrt{\left(\mathcal{B}(t) - \mathcal{A}(t)\right)^2 + \mathcal{C}(t)^2} \right) \\ \approx& \frac{4\mathcal{A}\mathcal{B} - \mathcal{C}^2}{4\mathcal{A}} \end{aligned} \tag{64}$$

so

$$\xi^2 = N\frac{4\mathcal{A}\mathcal{B} - \mathcal{C}^2}{4\mathcal{A}|\langle \mathbf{J} \rangle|^2}. \tag{65}$$

Now, since we are considering short squeezing times we have small $\chi_0 t$, and approximate $\cos(\chi_0 t) \approx \mathrm{e}^{-(\chi_0 t)^2/2}$. We also use large N to drop terms $S + y = S$ for half or whole small integer y.

$$\begin{aligned} \Delta J_y^2 &= \frac{S\mathrm{e}^{-\gamma_l t}}{2} + \frac{S^2}{2}\left(1 - \mathrm{e}^{-4S(\chi_0 t)^2)}\right)\mathrm{e}^{-2\gamma_l t} \\ \Delta J_z^2 &= \frac{S\mathrm{e}^{-\gamma_l t}}{2} \\ \langle \{\hat{J}_y, \hat{J}_z\} \rangle &= 2S^2\chi_0 t\mathrm{e}^{-S(\chi_0 t)^2}\mathrm{e}^{-2\gamma_l t}. \end{aligned} \tag{66}$$

We identify $q = S(\chi_0 t)^2$ as a small unitless parameter and series expand to $\mathcal{O}(q^3)$.

$$\begin{aligned} \Delta J_y^2 =& \frac{S\mathrm{e}^{-\gamma_l t}}{2} + \frac{S^2}{2}\left(1 - \mathrm{e}^{-4q}\right)\mathrm{e}^{-2\gamma_l t} \\ \approx& \frac{S\mathrm{e}^{-\gamma_l t}}{2} + 2S^2 q\left(1 - 2q + \frac{8}{3}q^2\right)\mathrm{e}^{-2\gamma_l t} \\ \langle \{\hat{J}_y, \hat{J}_z\} \rangle^2 =& 4S^3 q\mathrm{e}^{-2q}\mathrm{e}^{-4\gamma_l t} \\ \approx& 4S^3 q\mathrm{e}^{-4\gamma_l t}\left(1 - 2q + 2q^2\right). \end{aligned} \tag{67}$$

This all yields

$$\begin{aligned} 4\mathcal{A}\mathcal{B} - \mathcal{C}^2 =& S^2\mathrm{e}^{-2\gamma_l t} + 4S^3 q\left(1 - 2q + \frac{8}{3}q^2\right)\mathrm{e}^{-3\gamma_l t} \\ &- 4S^3 q\mathrm{e}^{-4\gamma_l t}\left(1 - 2q + 2q^2\right) \end{aligned} \tag{68}$$

and

$$\begin{aligned} \mathcal{A} =& \frac{S\mathrm{e}^{-\gamma_l t}}{2} + 2S^2 q\left(1 - 2q + \frac{8}{3}q^2\right)\mathrm{e}^{-2\gamma_l t} \\ \approx& 2S^2 q\mathrm{e}^{-2\gamma_l t}. \end{aligned} \tag{69}$$

Now, we collect terms and drop them according to $S^2 q \gg S$ for strongly squeezed quadratures, yielding

$$\begin{aligned} \xi^2 =& \frac{4\mathcal{A}\mathcal{B} - \mathcal{C}^2}{4\mathcal{A}|\langle \mathbf{J} \rangle|^2/N} \\ \approx& \frac{\mathrm{e}^{+2\gamma_l t}}{4Sq} + \left(\mathrm{e}^{+\gamma_l t} - 1\right) \end{aligned} \tag{70}$$

where, because we aren't near the limit of oversqueezed states, we don't have the typical curvature term $\propto q^2$. Lastly, we add back in the term for collective emission. In the limit of large N, we have that $V_- \approx \mathcal{B} = \Delta J_z^2$, and therefore

$$\xi^2 \approx (\xi_0)^2 + \frac{N(T(\Gamma_+) + T(\Gamma_-))}{|\langle \mathbf{J} \rangle|^2} \tag{71}$$

where $(\xi_0)^2$ is the spin squeezing parameter without collective decay. This yields

$$\begin{aligned} \xi^2 &\approx \frac{\mathrm{e}^{+2\gamma_l t}}{4Sq} + \left(\mathrm{e}^{+\gamma_l t} - 1\right) + \frac{N(T(\Gamma_+) + T(\Gamma_-))}{|\langle \mathbf{J} \rangle|^2} \\ &= \mathrm{e}^{2\gamma_l t}\left(\frac{1}{4Sq} + 2\frac{T(\Gamma_+) + T(\Gamma_-)}{S}\right) + \left(\mathrm{e}^{+\gamma_l t} - 1\right). \end{aligned} \tag{72}$$

Where, lastly, we series expand to first order in $\Gamma_\pm, \gamma_l$.

$$\xi^2 \approx \frac{1}{4Sq} + t\gamma_l + 4St\left(\Gamma_+ + \Gamma_-\right), \tag{73}$$

yielding, as a final expression,

$$\xi^2 = \frac{1}{(N\chi_0 t)^2} + t\gamma_l + 2Nt\Gamma_c, \tag{74}$$

where $\Gamma_c = \Gamma_+ + \Gamma_-$ is the collective decay defined in the main text.

As a reminder:

$$\begin{aligned} \gamma_l \equiv& \frac{\gamma U_0|\beta|^2}{2\Delta_a}, \quad \Gamma_\pm \equiv \frac{\kappa|\beta|^2|A^\pm|^2}{4}, \\ |\beta|^2 \equiv& \frac{\eta^2}{(\Delta_c')^2 + \frac{\kappa^2}{4}}, \quad A^\pm = \frac{U_0}{\Delta_c' \pm \omega_z - \frac{i\kappa}{2}}. \end{aligned} \tag{75}$$

When only collective decay is considered,

$$\frac{1}{(N\chi_0 t)^2}, 2Nt\Gamma_c \gg t\gamma_l \tag{76}$$

and one finds [8]

$$\xi^2 \approx 3\left(\frac{\Gamma_c}{2\chi}\right)^{2/3}, \quad t_{\mathrm{opt}} = \frac{1}{N(\chi\Gamma_c)^{1/3}}, \tag{77}$$

which is independent of N.

One can improve this by considering longer timescales and solving for the dressed cavity detuning such that the coopertivity is of order 1;

$$\gamma_l = 2N\Gamma_c \tag{78}$$

so we find

$$\Delta'_{c,\mathrm{opt}} \approx \pm\sqrt{\frac{2NU_0\Delta_a\kappa}{\gamma}}. \tag{79}$$

where the positive or negative solutions correspond to picking the peak from the red or blue shifted sidebands, A^+ or A^-, respectively.

This yields

$$\xi^2 = 2\gamma_l t + \frac{1}{(N\chi_0 t)^2} \tag{80}$$

which we optimize with respect to t to find

$$t_{\mathrm{opt}} = \left(\gamma_l\ (N\chi_0)^2\right)^{-1/3}, \tag{81}$$

which, to highest order in N, is

$$t_{\mathrm{opt}} \approx 4\left(\frac{2N^2\Delta_a^5\kappa^4}{U_0\gamma^5\eta^6}\right)^{1/3} \tag{82}$$

giving

$$\xi^2 \approx 3\left(\frac{2\gamma\kappa}{NU_0\Delta_a}\right)^{1/3}. \tag{83}$$

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